Math 340: Elementary Matrix and Linear Algebra

MIDTERM ONE

Thursday, Feburary 27th, 2024, 7:30pm-9:00pm

Circle your Instructor and your TA:

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I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: ______________________________

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of seven problems for a total of 50 points; the exam is six sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam packet will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- If making multiple attempts at a solution, indicate clearly which attempt you'd like to be graded by crossing out the other answers. Multiple attempts will not be graded.
- When doing multiple choice and true/false questions, make sure to fully fill in the chosen bubble.

  Correct mark: ● Incorrect marks: ◐ ⊗ ⊘ ✓
- **Notation:** $\mathbf{0}$ denotes the zero vector, O denotes the zero matrix, and I_n denotes the $n \times n$ identity matrix.

DO NOT BEGIN THIS EXAM UNTIL SIGNALED TO DO SO.

1. (8 points total) Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ by $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 3x_1 - x_2 + 2x_3 \\ 2x_1 \\ 2x_1 + 4x_3 \end{bmatrix}$.

a. (2 points) Evaluate $T\left(\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}\right)$. $= \begin{bmatrix} 3\cdot 3 - 1 + 2\cdot 2 \\ 2\cdot 3 \\ 2\cdot 3 + 4\cdot 2 \end{bmatrix} = \begin{bmatrix} 12 \\ 6 \\ 14 \end{bmatrix}$

b. (2 points) Find the (induced) matrix A such that $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^3$.

$$A = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 4 \end{bmatrix}$$

c. (4 points) Is there a **nonzero** vector $\mathbf{x}$ in $\mathbb{R}^3$ such that $T(\mathbf{x}) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$? Fully justify your answer.

① $\det(A) = -2(-4-2) = 12 \neq 0$

No. since A is invertible, the only solution to $A\vec{x} = \vec{0}$ is the trivial solution, $\vec{x} = \vec{0}$.

② Solve: $3x_1 - x_2 + 2x_3 = 0 \Rightarrow 0 - x_2 + 0 = 0 \Rightarrow x_2 = 0$

$2x_1 = 0 \Rightarrow x_1 = 0$

$2x_1 + 4x_3 = 0 \Rightarrow 0 + 4x_3 = 0 \Rightarrow x_3 = 0$

No, only soln. is $x_1 = 0$, $x_2 = 0$, $x_3 = 0$

③ $$A = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 4 \end{bmatrix} \xrightarrow[r_1 \leftrightarrow r_2]{\frac{1}{2}r_2 \to r_2 \text{ and}} \begin{bmatrix} 1 & 0 & 0 \\ 3 & -1 & 2 \\ 2 & 0 & 4 \end{bmatrix} \xrightarrow[-2r_1 + r_3 \to r_3]{-3r_1 + r_2 \to r_2} \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 4 \end{bmatrix} \end{array}$$

$\Rightarrow x_1 = 0$

$\Rightarrow -x_2 + 2x_3 = 0$

$\Rightarrow 4x_3 = 0$

$\Downarrow$

$x_3 = 0$

$\Downarrow$

$x_2 = 0$

2. (8 points total) Suppose A is a 3×3 matrix and $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ is an eigenvector of A corresponding to eigenvalue -2, and $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$ is an eigenvector of A corresponding to eigenvalue 3. Compute each of the following (note: your final answers will be vectors in $\mathbb{R}^3$). Show all work.

a. (2 points)

$$A \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix} = A \cdot 2\vec{v} = 2 \cdot 3\vec{v} = \begin{bmatrix} 12 \\ 6 \\ 6 \end{bmatrix}$$

b. (2 points)

$$A^3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = A^2 A\vec{u} = A^2(-2)\vec{u} = -2AA\vec{u} = (-2)^2 A\vec{u} = (-2)^3\vec{u}$$

$$= \begin{bmatrix} -8 \\ 8 \\ -8 \end{bmatrix}$$

c. (4 points)

$$A \begin{bmatrix} -1 \\ -5 \\ 1 \end{bmatrix} = A(3\vec{u} - 2\vec{v}) = 3A\vec{u} - 2A\vec{v} = \begin{bmatrix} -6 \\ 6 \\ -6 \end{bmatrix} + \begin{bmatrix} -12 \\ -6 \\ -6 \end{bmatrix} = \begin{bmatrix} -18 \\ 0 \\ -12 \end{bmatrix}$$

(hint: Can you write this vector as a linear combination of **u** and **v**?)

$$\overset{3}{a}\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + \overset{-2}{b}\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \\ 1 \end{bmatrix}$$

$$a + 2b = -1$$
$$+ \; -a + b = -5$$
$$3b = -6 \Rightarrow b = -2$$
$$a = -1 - 2(-2) = 3$$

Check: $3 + -2 = 1$ ✓.

3. (8 points, 1 point each) Clearly mark the correct answer for each of the following by **completely** filling in the appropriate bubble. **No justification is needed.**

   a. **(True/False)** If A and B are both 3×3 matrices, then $(A+B)^2 = A^2 + 2AB + B^2$.

   - ○ True
   - ● False

   $$(A+B)^2 = (A+B)(A+B) = A^2 + \underbrace{AB + BA}_{\text{may not equal } 2AB.} + B^2$$

   b. **(True/False)** $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} xy \\ 0 \end{bmatrix}$ is a linear transformation.

   - ○ True
   - ● False

   c. **(True/False)** The matrix $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ is a reduced row echelon form matrix.

   - ● True
   - ○ False

   d. **(True/False)** If A is a 2×2 matrix such that for every 2×2 matrix B, we have $AB = BA$, then A must be I_2.

   - ○ True
   - ● False

   ex: $A = 0$.

   e. **(True/False)** If $T : \mathbb{R}^5 \to \mathbb{R}^4$ is a matrix transformation defined by $T(\mathbf{x}) = A\mathbf{x}$, then A is a 5×4 matrix.

   - ○ True
   - ● False

   $\vec{x}$ is 5×1
   $A\vec{x}$ is 4×1
   $\Rightarrow A$ is 4×5

f. **(Multiple Choice-choose one)** For $M = \begin{bmatrix} 2 & 3 & 1 \\ -3 & c & 2 \\ 1 & 0 & 2 \end{bmatrix}$, for which values(s) of c would M be **not invertible**?

- ○ $c = \frac{12}{5}$
- ● $c = -8$
- ○ All values of c except $c = -8$
- ○ $c = 0$
- ○ None of the above.

$\to$

$1(6-c)+2(2c+9)=0$

$24+3c=0$

$c=-8$

g. **(Multiple Choice-choose one)** What are the eigenvalues of the matrix $A = \begin{bmatrix} -2 & 1 \\ 3 & 0 \end{bmatrix}$?

- ○ $\lambda = 0, 2$
- ○ $\lambda = 1, 3$
- ○ $\lambda = -1, 3$
- ● $\lambda = -3, 1$
- ○ None of the above.

$\det(\lambda I - A) = \det \begin{bmatrix} \lambda+2 & -1 \\ -3 & \lambda \end{bmatrix}$

$= (\lambda+2)(\lambda) - 3 = 0$

$\lambda^2 + 2\lambda - 3 = 0$

$(\lambda+3)(\lambda-1) = 0$

$\lambda = -3, 1$

h. **(Multiple Choice-choose one)**

For which s and t is the matrix $M = \begin{bmatrix} 0 & t & s \\ 3t & 0 & s+t \\ 4 & 4 & t \end{bmatrix}$ **symmetric**?

- ○ $s = 0, t = 0$
- ○ $s = -4, t = 0$
- ● $s = 4, t = 0$
- ○ $s = 0$, t can be any value.
- ○ None of the above.

$3t = t \Rightarrow t = 0$

$s = 4 \quad 4+0=4\checkmark.$

4. (8 points total)

a. (4 points) Suppose A and B are 4×4 matrices with $2A^T B^{-1} + I_4 = O$. If $\det(A) = 10$, find the determinant of B.

$$2A^T B^{-1} = -I_4 \longrightarrow \det(2A^T B^{-1}) = \det(-I_4) = 1$$

$$2^4 \det(A^T) \frac{1}{\det(B)} = 1$$

$$2^4 \cdot 10 \cdot \frac{1}{\det(B)} = 1$$

$$\Rightarrow \boxed{\det(B) = 2^4 \cdot 10}$$

b. (4 points) Suppose C, D are 4×4 matrices, and D is obtained from C by the following elementary row operations:

C

- Add $3r_2$ to r_4 (and replace r_4) nothing
- Multiply r_3 by k (for some unknown scalar k) $k\det(C)$
- Swap r_3 and r_4 $-k\det(C)$
- Add $-r_1$ to r_2 (and replace r_2) nothing
- Multiply r_2 by 5 $-5k\det(C)$

D

If $\det(C) = 10$ and $\det(D) = -4$, find the scalar k. Show all work.

$$\det(D) = -4 = -5k \cdot 10$$

$$\boxed{k = \frac{-4}{-5 \cdot 10}}$$

5. (6 points total) Suppose a system of three equations in four unknowns (x_1, x_2, x_3, x_4) is represented by the follow augmented matrix:

$$\left[\begin{array}{cccc:c} 1 & 2 & 1 & -1 & 3 \\ -1 & -2 & 1 & -3 & 5 \\ 2 & 4 & 2 & -2 & 6 \end{array}\right]$$

a. (2 points) Write out the system of linear equations corresponding to this augmented matrix.

$x_1 + 2x_2 + x_3 - x_4 = 3$

$-x_1 - 2x_2 + x_3 - 3x_4 = 5$

$2x_1 + 4x_2 + 2x_3 - 2x_4 = 6$

b. (4 points) Through elementary row operations, we have the following reduction to reduced row echelon form:

$$\left[\begin{array}{cccc:c} 1 & 2 & 1 & -1 & 3 \\ -1 & -2 & 1 & -3 & 5 \\ 2 & 4 & 2 & -2 & 6 \end{array}\right] \to \left[\begin{array}{cccc:c} 1 & 2 & 0 & 1 & -1 \\ 0 & 0 & 1 & -2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Using the reduced row echelon form, write out the parametrized solution for x_1, x_2, x_3, x_4 for this linear system.

$x_2 = t$ $\longrightarrow$ $x_1 = -1 - 2t - s$

$x_4 = s$ $\qquad x_2 = t$

$x_1 + 2x_2 + x_4 = -1$ $\qquad x_3 = 4 + 2s$

$x_3 - 2x_4 = 4$ $\qquad x_4 = s$

6. (8 points total)

Consider 2×2 matrices A, B, C, D, where $B = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix}$, and $D = \begin{bmatrix} 3 & 4 \\ -1 & -1 \end{bmatrix}$.

a. (3 points) Find D^{-1}. Show all work.

$$D^{-1} = \frac{1}{\underbrace{(3)(-1)-(-1)(4)}_{=1}} \begin{bmatrix} -1 & -4 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} -1 & -4 \\ 1 & 3 \end{bmatrix}$$

or

$$\left[\begin{array}{cc|cc} 3 & 4 & 1 & 0 \\ -1 & -1 & 0 & 1 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_2} \left[\begin{array}{cc|cc} -1 & -1 & 0 & 1 \\ 3 & 4 & 1 & 0 \end{array}\right] \xrightarrow{-r_1 \to r_1} \left[\begin{array}{cc|cc} 1 & 1 & 0 & -1 \\ 3 & 4 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{-3r_1 + r_2 \to r_2} \left[\begin{array}{cc|cc} 1 & 1 & 0 & -1 \\ 0 & 1 & 1 & 3 \end{array}\right] \xrightarrow{-r_2 + r_1 \to r_1} \left[\begin{array}{cc|cc} 1 & 0 & -1 & -4 \\ 0 & 1 & 1 & 3 \end{array}\right] \rightsquigarrow D^{-1} = \begin{bmatrix} -1 & -4 \\ 1 & 3 \end{bmatrix}$$

b. (5 points) If $(2A + B)^{-1} C = D$, find A. Show all work. (Hint: You can use your answer to part (a)).

$\downarrow$ mult. by $(2A+B)$

$$C = (2A + B)D$$

$\downarrow$ mult. by D^{-1}

$$CD^{-1} = 2A + B$$

$$CD^{-1} = \begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} -1 & -4 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} -2 & -9 \\ -3 & -11 \end{bmatrix}$$

$$\begin{bmatrix} -2 & -9 \\ -3 & -11 \end{bmatrix} = 2A + B \rightsquigarrow 2A = \begin{bmatrix} -2 & -9 \\ -3 & -11 \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} -4 & -8 \\ -4 & -14 \end{bmatrix}$$

$$\boxed{A = \begin{bmatrix} -2 & -4 \\ -2 & -7 \end{bmatrix}}$$

7. (4 points total) Suppose A is a 4×4 matrix that is row-equivalent to the matrix $B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Answer each of the following questions about the matrix A, if possible. Justification is not required. (Another reminder: You are answering questions about A, not B).

a. **(Multiple Choice-choose one)** Is A invertible? No, it reduces to a matrix that has a row of zeros.

- ○ Yes
- ● No
- ○ Not enough information to answer.

b. **(Multiple Choice-choose one)** What is the rank of A?

same as rank of B, which is 3 = # of leading ones.

- ○ 0
- ○ 1
- ○ 2
- ● 3
- ○ 4
- ○ Not enough information to answer.

c. **(Multiple Choice-choose one)** For $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}$, does the linear system $A\mathbf{x} = \mathbf{b}$ have...

could have no solutions <u>or</u> infinitely many. depends on original A matrix. (see next page)

- ○ No solutions
- ○ One solution
- ○ Infinitely many solutions
- ● Not enough information to answer.

d. **(Multiple Choice-choose one)** For $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, does the linear system $A\mathbf{x} = \mathbf{b}$ have...

must be consistent w/ free variables.

- ○ No solutions
- ○ One solution
- ● Infinitely many solutions
- ○ Not enough information to answer.

Scratch paper page!!

$\boxed{7c}$

ex:

inf. many

$-2r_1+r_2\to r_2$, $r_1+r_3\to r_3$, $-3r_1+r_4\to r_4$

$$\left[\begin{array}{c|c} \underbrace{\begin{matrix}1 & 2 & 3 & 4\\ 0 & 1 & 3 & 0\\ 0 & 0 & 1 & -4\\ 0 & 0 & 0 & 0\end{matrix}}_{B} & \begin{matrix}1\\0\\0\\0\end{matrix}\end{array}\right] \xrightarrow[3r_1+r_4\to r_4]{\substack{2r_1+r_2\to r_2\\ -r_1+r_3\to r_3}} \left[\begin{array}{c|c} \underbrace{\begin{matrix}1 & 2 & 3 & 4\\ 2 & 5 & 9 & 8\\ -1 & -2 & -2 & -8\\ 3 & 6 & 9 & 12\end{matrix}}_{A} & \begin{matrix}1\\2\\-1\\3\end{matrix}\end{array}\right]$$

For this A, $A\vec{x}=\begin{bmatrix}1\\2\\-1\\3\end{bmatrix}$ has infinitely many solutions.

ex:

no solutions

If $A=B$, $A\vec{x}=\begin{bmatrix}1\\2\\-1\\3\end{bmatrix}$ has no solutions.

Another Scratch paper page!!

Back of scratch paper page!!